Ch- 1 Human Eye and the Colourful World

• Parts of human eye-

1 Cornea: It is the outermost portion of the eye. It is transparent.
Function- most of the refraction of light occurs at cornea.

2 Iris: It is circular brown structure present in the eye. In some people it is light-blue in colour.
Function- It controls the size of pupil.

3 Pupil: It is a black circular dome present on the iris.
Function- It controls the amount of light entering into the eye.

4 Aqueous humour: It is a jelly-like structure present on the cornea. It maintains the intra-ocular pressure of the human eye.
Function- maintains intra-ocular pressure.

5 Eye lens: It is crystalline in nature. It is a convex lens. It is present just behind pupil.
Function- It helps in image formation on the retina.

6 Ciliary Muscles: It is a fibrous tissue attached to the end of eye lens.
Function- It helps in elongation and contraction of eye lens.

7 Vitrous humour: It is a jell-like structure present in the human eye.
Function- It gives shape to the eyeball.

8 Retina: It is a point present in the human eye.
Function- Image is formed on the retina. It consists of Rod and Cone cells.
* Rod cells: It is active when intensity of light is dim.
* Cone cells: It is active when intensity of light is bright. It also helps to see colours.

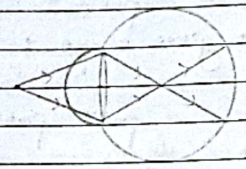
9 Optic nerve: It converts image formed in the retina into electrical impulses and transmit it to the human brain.

Scanned with OKEN Scanner

• Defects of Vision-

1 Myopia (Short-sightedness): In this defect, a person is able to see the nearby objects clearly but is not able to see distant objects clearly.

Defective lens



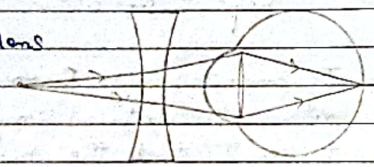
* Cause of defect-

a Elongation of Eye-ball

b Excessive curvature of eye-lens

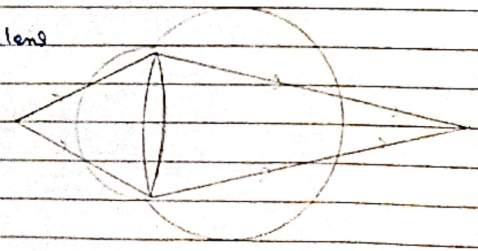
* Correction of defect-

Concave lens



2 Hypermetropia (long-sightedness). A person is able to see distant objects clearly but can't see nearby objects distinctly.

Defective lens



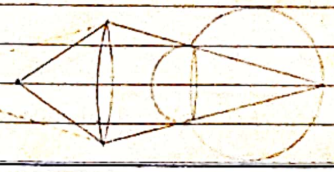
* Cause of defect-

1 The focus length of the eye is too long.

2 The eye ball has became too small.

* Correction of defect-

Convex lens



• Power of Accomadation-

It is the ability of a eye lens to adjust its focal length so that the object can be seen clearly.

- The least distance of distinct vision for a normal human eye is 25 cm.

Myopia (Concave lens)	Hypermetropia (convex lens)
u = infinity (∞)	u = 25 cm
v = -ve	v = -ve
f = -ve	f = +ve
P = -ve	P = +ve

Scanned with OKEN Scanner

Questions

Q1 The far point of a defective eye is 80 cm

(i) Identify the defect

(ii) State two cases of defect

(iii) Corrective power of lens required

(i) - Ans Myopia

(ii) - Ans Elongation of eye ball; Excessive curvature

(iii) - Ans

$$P = \frac{100}{f}$$

$$= \frac{\cancel{100}^{5}}{\cancel{480}_{4}} = \frac{5}{4} = 1.25\,D$$

$$P = \frac{1}{f} = \frac{1}{\frac{1}{3}} = 3D$$

Focal length was positive, it is a convex lens.

Q8 Why ---------- 25 cm?

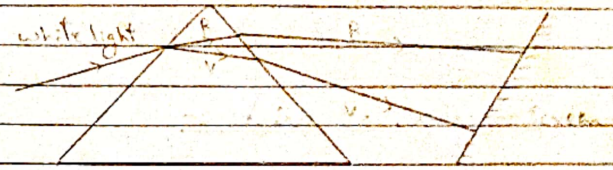
Ans At distance less than 25 cm, the ciliary muscles cannot bulge the eye lens any more, the object cannot be focused on the retina and it appears blurred to the eye.

Q9 What ---------- eye?

Ans The eye lens of a normal eye forms the images of objects of various distances on the same retina. Thus, the image distance in the eye remains the same.

- **Dispersion of Light -**

- Splitting of white light into its constituent colours on passing through a glass prism.



Decrease in wavelength		R O Y G B I V	Increase in wavelength

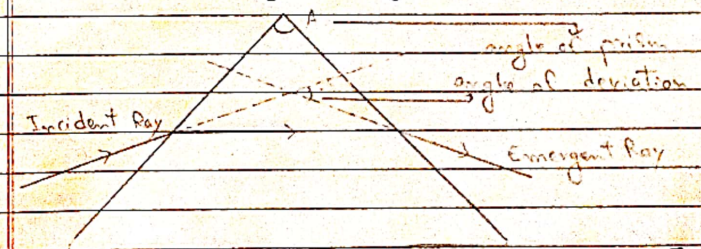
Distance = v × time

wavelength = v × time period

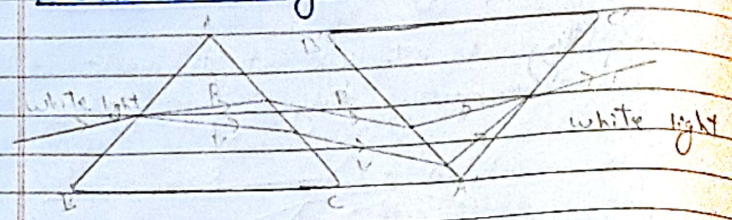
$$\lambda = v \times T$$

$$\lambda = \frac{v}{\nu} \qquad \left[T = \frac{1}{\nu}\right]$$

- **Refraction of Light through a prism -**

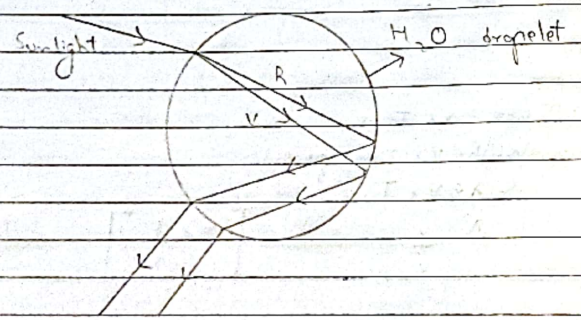


- **Recombination of Light-**



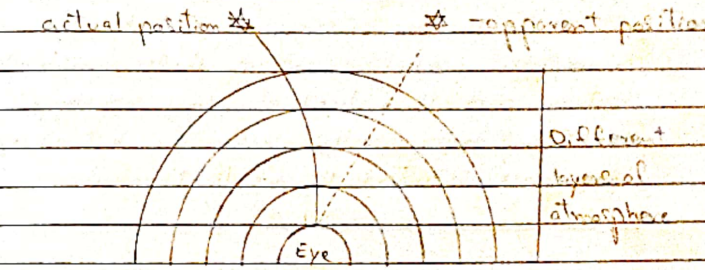
- **Formation of Rainbow-**

* Conditions-
  - It is always formed when the sun comes out just after rain.
  - It is always formed opposite to the sun.
  - The water dropelets present in the atmosphere acts as a glass prism.



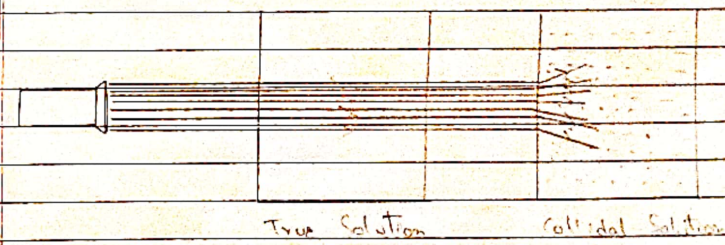
- **Twinkling of light-**

- Stars don't actually Twinkle, they appear to twinkle.
- Starlight acts as a point source of light.
- when starlight passes through different layers of atmosphere, it undergoes multiple atmospheric refraction.



- **Scattering of Light-**

- Deviation of light from its original path & spreading into the region where light passes through a collidal solution.



Scanned with OKEN Scanner

- **Tyndall Effect -**

- It is a practical application of scattering of light.
- When white light passes through a canopy (a small opening), light scatters making the particles present in its path visible.

Q1 Why is the colour of clear sky blue?

Ans) It is due to the scattering of light by the atmospheric particles. Light of shorter wavelength scatters more than light of longer wavelength. Voilet has more shorter wavelength, but our eyes are more sensitive to blue light.

Scanned with OKEN Scanner